SYNCOTEK
Hardware Solution Provider

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Card Issuing Machine

Model: SK-AD3

REV: A 2011-12-03

Revision History

REV.	DATE	ENGINEER	DESCRIPTION
A	2011-12-03	Mr Liu	Newly Issued

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1 Product Image



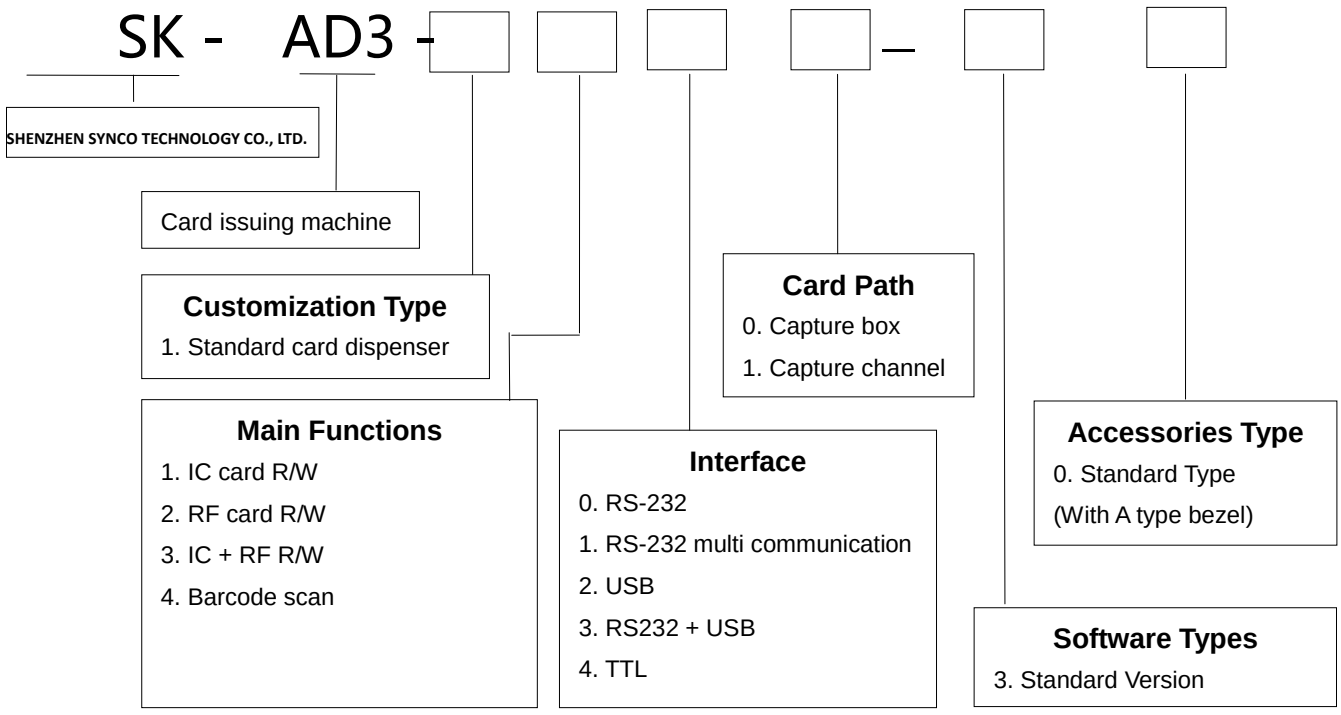
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2 Overview

SK-AD3 card issuing machine supports IC/RF card R/W and issuing/collecting functions, which is primarily applicable to self-service terminal equipments such as parking lot management system, expressway toll gate, self-service charging system, self-service card vending terminal, and electricity bill payment system.

3 Model Number



C/D: Card Dispenser Series

R/W: READ&WRITE

Accessories options:

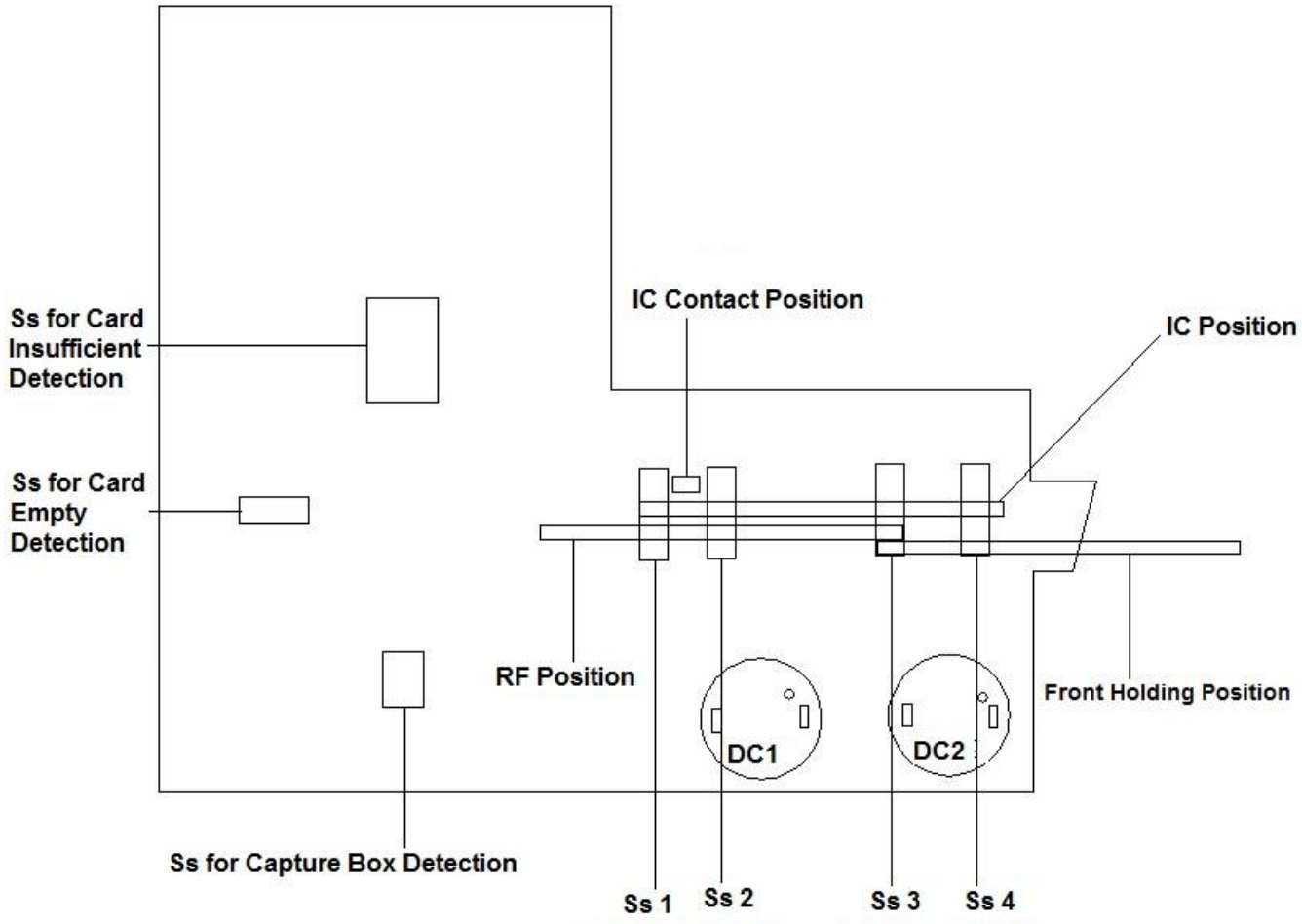
*Stacker (model no.: S003)

*Barcode scanning module

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4 Structure and Card Stop Position



Ss: Sensor

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5 Features

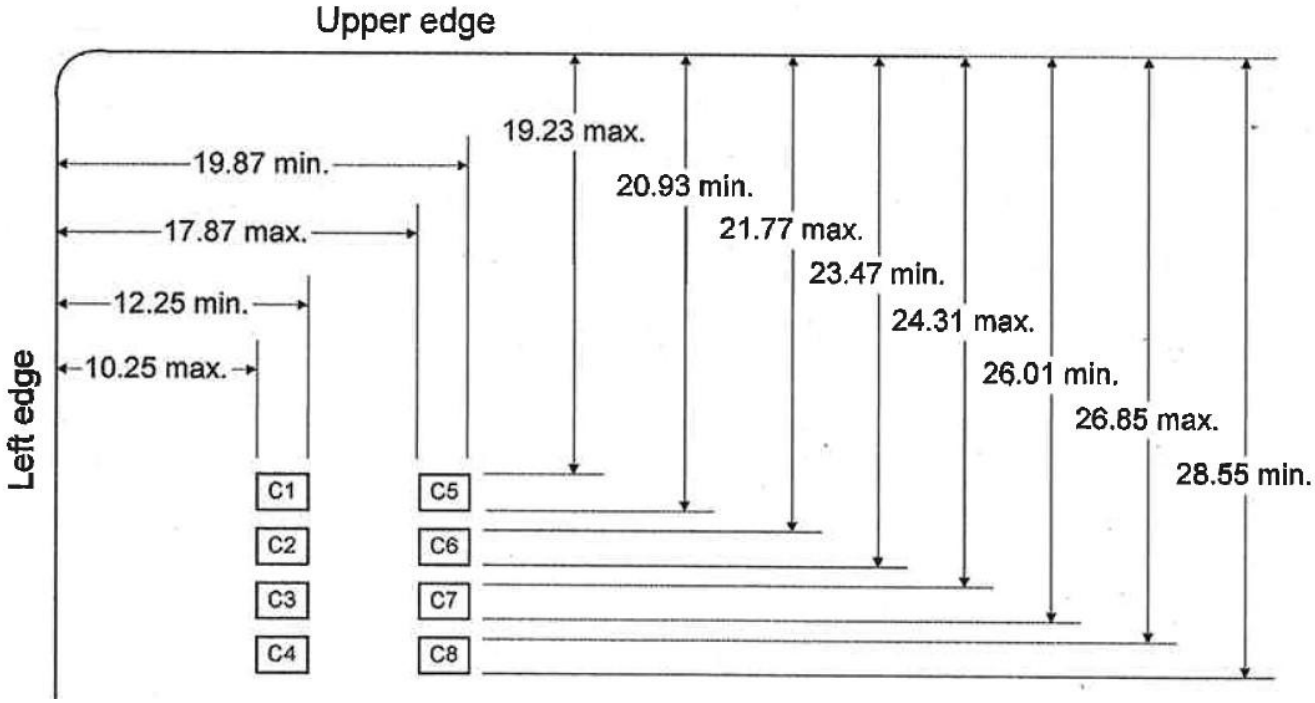
- Supports ISO7816-2 contact type IC card reading/writing.
- Supports ISO14443-A contactless Type A/B card reading/writing.
- Unique dispensing wheel design: is capable of issuing various embossing cards accurately.
- Automatic internal adjustment: ensures accurate issuing cards of different thicknesses.
- Multi functions in one device: provide IC/RF card reading/writing, issuing/collecting functions and effectively saves space for the complete machine.
- Rear card capture box: provides convenience for card management.
- Simple collection channel design: satisfies user requirement for large collection amount.
- Multiple sensors: are able to judge the card position and indicate the card status.
- Scientific card transfer wheel design: automatically avoids dust accumulation which will lead to slippage.
- High-strength wearable plastic channel: is adopted to reduce the risk of metal shielding and increase the success rate of RF card reading/writing.
- Multiple levels of sensor circuit protection: prevents optical interference.
- PCSC protocol: is available.

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6 Technical Specifications

6.1 Applicable Contact IC Card

ISO7816, referring to electronic identification cards with contacts, especially smart cards, describes eight contacts from C1 to C8, of which dimension is shown as below left and upper edges of the card.



(Unit: mm)

6.2 Applicable Contactless IC Card

Contactless IC card supported conforms to ISO14443 international standard Type A/B card..

Notes: There are totally 16 sectors for 1 S50/S70 and each sector has an independent operational access key.

ITEM	CONTENT
Standard	ISO 14443
Operating Frequency	13.56MHz
Communication Speed	105 bps

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6.3 Card Supported

86L x 55W mm (0.2~1.2mm thickness; factory standard: 0.8mm)

6.4 Communication Method

USB (PC/SC)/ RS232 (optional)
(RS232 supports multi communication with default address 00.)

6.5 Operating Conditions

Power Supply	DC24V±5%, 2.5A
Quiescent Current	200mA
Default Baud Rate	9600BPS/sec.

6.6 Card Stacker Capacity

130 pcs (0.76mm embossing card)
175 pcs (0.76mm standard card)
350 pcs (0.4mm thickness)

6.7 Capture Box Capacity

25 pcs (0.76mm standard card)

6.8 Service Life

Issuing Life	1,000,000 passes min.
IC Contact	500,000 passes min.

6.9 Operation / Storage Environment

Operation Temperature / Humidity	0°C to 50°C, 0 to 90% RH without condensation
Storage Temperature / Humidity	-25°C to 80°C, 0 to 90% RH without condensation

6.10 Net Weight

Approx. 1.6KG (excluding accessories and package)

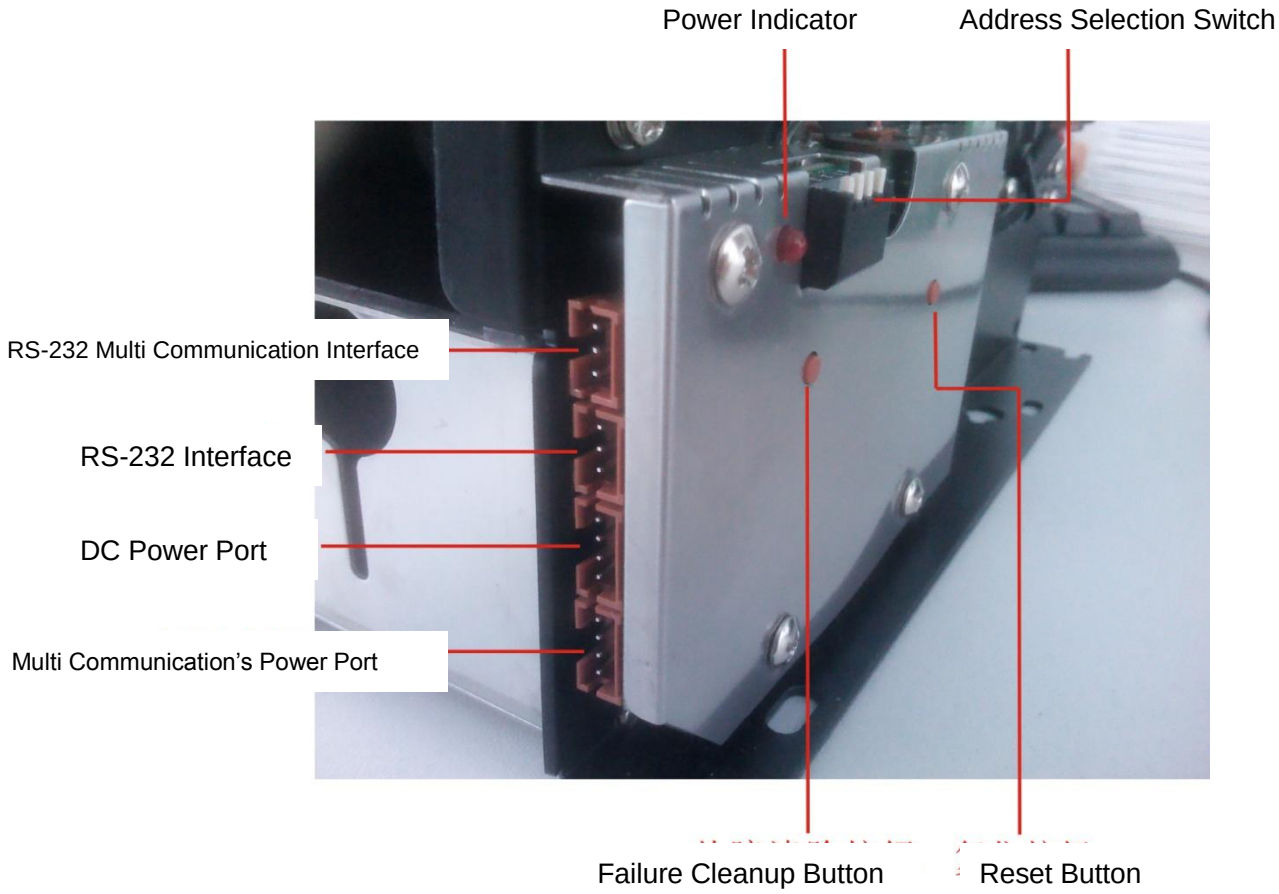
6.11 Dimension

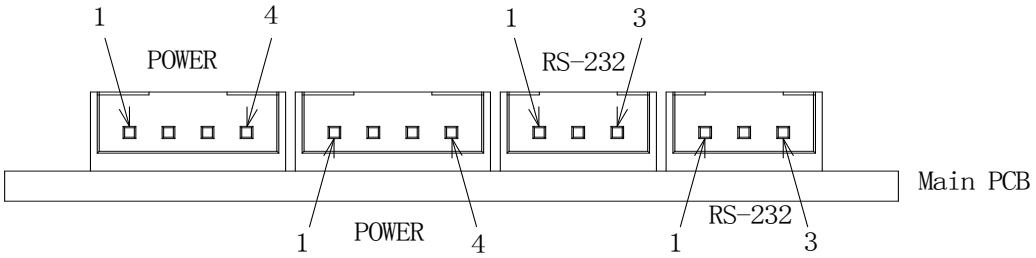
233L x 69W x 200H mm

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6.12 RS232 Interface Descriptions

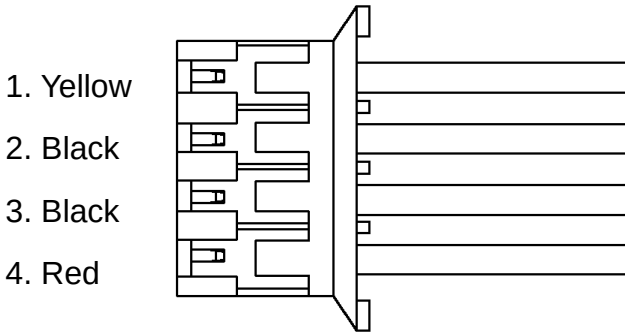




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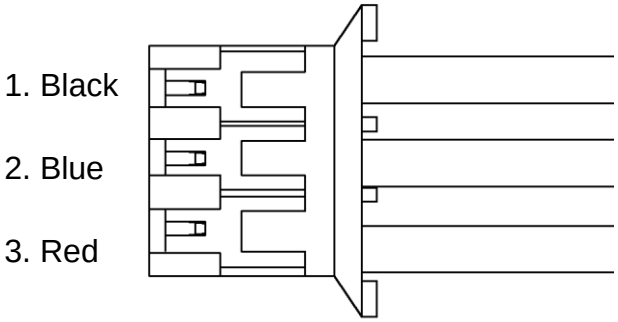
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6.13 Power port descriptions



No.	Signal	Functions
1	BPS	Backup power supply (Yellow)
2	GND	Ground (Black)
3	GND	Ground (Black)
4	+24V	DC+24V power input port (Red)

6.14 Com port descriptions



No.	Signal	Functions
1	SGN	Ground switch (Black)
2	TXD	PC transmit data (Blue)
3	RXD	PC receive data (Red)

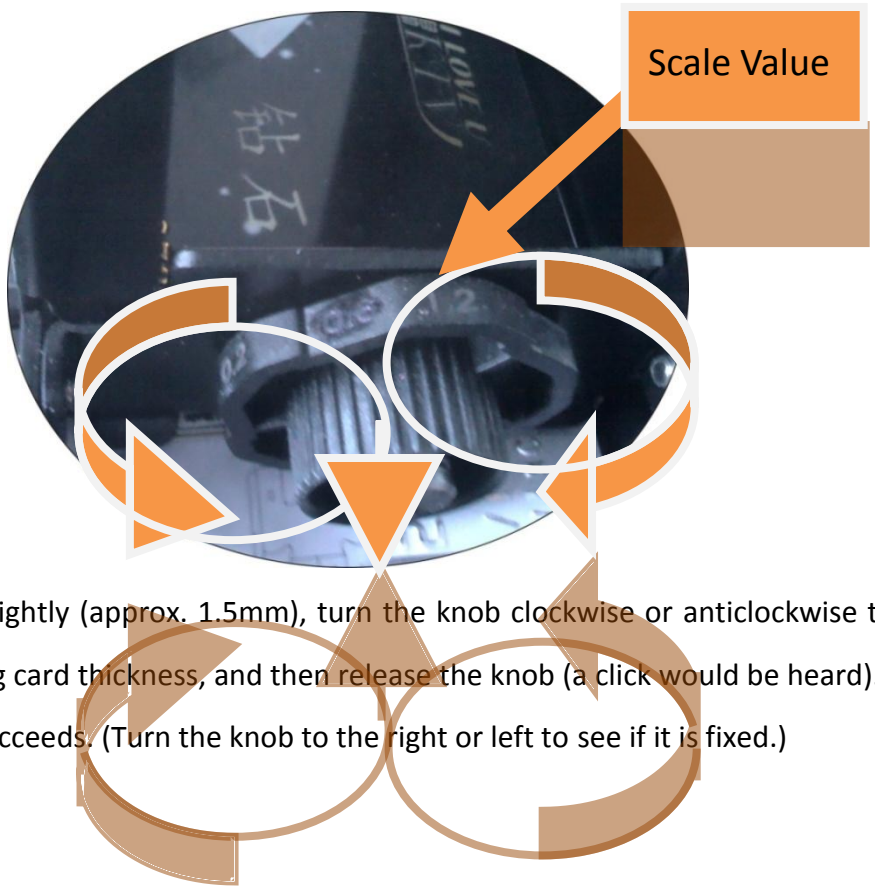
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6.15 Operation sequence

7.3.1. Connect power line and RS232 serial port line (pay attention to the negative and positive).

7.3.2. Open DEMO and select corresponding COM NO. to set baud rate. Baud rate 1200~1152000BPS could be selected and default is 9600BPS. Click OPEN COM in DEMO and program will automatically search for COM NO. and baud rate.

7.3.3. Card thickness adjustment



Pull out the turn-knob lightly (approx. 1.5mm), turn the knob clockwise or anticlockwise to the scale value of dispensing card thickness, and then release the knob (a click would be heard). Card thickness adjustment succeeds. (Turn the knob to the right or left to see if it is fixed.)

7 Maintenance

If C/D has dispensed 50,000 cards or worked for 5 months indoor, below maintenance is highly recommended. (In the other case, once 3 months' maintenance is advised for outdoor application varying from the working environment.)

1. Press the failure cleanup button when no card is inside, and wipe the rubbing wheel in the bottom of

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dispenser, rubber roller in card dispensing channel, and IC contact with soft cloth with alcohol to clear foreign materials.

2. Clean sensors in card dispensing channel with a small brush or dust blower.

3. Check the tightness and wear of belts to do adjustment or change if needed.

4. Check the gap of shutter and insert two cards together, and if two cards insertion succeeds, please adjust the turn-knob to accurate scale value.

Notes:

One year's free warranty is provided started from the date when client received the goods, while damage caused by human reason is not within the warranty.

For following cases, fees for replaced parts or service charge would be required:

1) Mechanical damages caused by clients during the transportation or in the process of using.
2) Components or parts expired the guarantee time.
3) Damages caused by improper operations, or not conformed to user manual or warnings.
4) Breakdowns caused by personal disassembly, repair or modification.

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8 Dimensions

